

ARCHAEOLOGY



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the early history of the Earth, into the world of fossils. A paleontologist is a scientist who studies fossils of ancient organisms – they matter to us because they show how life on Earth has changed. Fossils are dug out of rock using hammers, chisels and brushes. The word “dinosaur” means “terrible lizard” in Greek. In the 19th century came the “Bone Wars,” a rivalry between Cope and Marsh. The most famous Tyrannosaurus skeleton, “Sue,” is in a museum in Chicago. Zofia Kielan-Jaworowska led Polish-Mongolian expeditions to the Gobi Desert, and in Krasiejów, in the Opole region, remains of reptiles about 230 million years old were found. Crowds fell in love with dinosaurs thanks to the 1993 film “Jurassic Park,” whose music was composed by John Williams. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The film “Jurassic Park” (1993) is famous for its music, composed by...
 - a) Ennio Morricone
 - b) Hans Zimmer
 - c) Wojciech Kilar
 - d) John Williams
2. 👁 The most famous Tyrannosaurus skeleton, “Sue”, is in a museum in...
 - a) Tokyo
 - b) Paris
 - c) Warsaw
 - d) Chicago
3. 🗨 In Greek, the word “dinosaur” means...
 - a) great bird
 - b) strong animal
 - c) old dragon
 - d) terrible lizard
4. 🏛 A paleontologist is a scientist who studies...
 - a) ruins of cities
 - b) fossils of ancient organisms
 - c) ancient languages
 - d) coins
5. 🔍 Fossils are extracted from rock using...
 - a) hammers, chisels and brushes
 - b) fire
 - c) only excavators
 - d) water from a hose
6. 🧑 The “Bone Wars” (19th century) were a rivalry between two paleontologists:
 - a) Cuvier and Lamarck
 - b) Cope and Marsh
 - c) Owen and Huxley
 - d) Darwin and Wallace
7. ❤ Zofia Kielan-Jaworowska led famous Polish–Mongolian expeditions to the...
 - a) Sahara Desert
 - b) Gobi Desert
 - c) Atacama Desert
 - d) Kalahari Desert
8. 🌿 In Krasiejów, in the Opole region of Poland, remains of reptiles were found that are about...
 - a) 23 years old
 - b) 2 billion years old
 - c) 230,000 years old
 - d) 230 million years old
9. ⭐ Fossils are important to us because they...
 - a) look nice
 - b) glow in the dark
 - c) show how life on Earth has changed
 - d) are expensive

PHILOSOPHY



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Let's travel back to the early history of the Earth and hear what philosophers thought about it. In the 18th century James Hutton wrote that in Earth's history "we find no vestige of a beginning, no prospect of an end." So the idea of "deep time" was born – the Earth is billions of years old. Georges Cuvier proved that species can go extinct, and the great extinctions show that life is fragile but recovers. Charles Darwin saw "grandeur in this view" because from a simple beginning endless forms most beautiful have evolved. Saint Augustine held that the biblical account of creation need not be read literally, like a science textbook. The Jesuit Teilhard de Chardin took part in the discovery of Peking Man. James Lovelock's Gaia hypothesis describes the Earth as a self-regulating system. And the thought of millions of years before humans leads to humility and wonder. Remember these thoughts – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 In the 18th century, James Hutton wrote that in Earth's history "we find no vestige of a beginning, no prospect of an..."
 - a) sea
 - b) people
 - c) song
 - d) end
2. 👁 The concept of "deep time" means that the Earth...
 - a) is billions of years old
 - b) is 100 years old
 - c) is eternal
 - d) is a few thousand years old
3. 🗨 James Lovelock's Gaia hypothesis describes the Earth as...
 - a) a machine
 - b) a self-regulating system
 - c) a flat disk
 - d) literally a giant animal
4. ➡ Georges Cuvier proved that species can...
 - a) arise from mud
 - b) go extinct
 - c) turn into stones
 - d) live forever
5. ✚ St. Augustine and other thinkers believed that the Bible's account of creation does not need to be read...
 - a) in Latin
 - b) at all
 - c) literally, like a science textbook
 - d) aloud
6. 🧑🏫 Teilhard de Chardin, a Jesuit and paleontologist, took part in the discovery of...
 - a) Ötzi
 - b) Peking Man
 - c) Tyrannosaurus
 - d) Lucy
7. ❤ Thinking about the millions of years before humans leads many people to...
 - a) fear of dinosaurs
 - b) boredom
 - c) humility and wonder
 - d) anger
8. 🌿 Charles Darwin wrote that there is "grandeur in this view of life", because from a simple beginning there evolved...
 - a) machines
 - b) endless forms most beautiful
 - c) cities
 - d) money
9. ⭐ The great mass extinctions show that life on Earth...
 - a) is fragile but recovers
 - b) never changes
 - c) does not matter
 - d) has always depended on humans

HISTORY

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the early history of the Earth and meet the people who uncovered it. Mary Anning, a pioneer of paleontology, is the heroine of the tongue twister “She sells seashells by the seashore.” In 1824 the first dinosaur received a scientific name – it was *Megalosaurus*. In 1830 Charles Lyell argued that the Earth was shaped by slow processes still at work today. The age of the Earth, about 4.5 billion years, was determined in 1956 by Clair Patterson, who also discovered that leaded gasoline poisons people and the environment. The first plants moved onto land about 470 million years ago. Dinosaurs appeared about 230 million years ago and died out about 66 million years ago – so they existed for 164 million years. In 1980 Luis and Walter Alvarez linked their extinction to an asteroid impact, whose trace is the Chicxulub crater in Mexico. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 **Mary Anning, a pioneer of paleontology, is said to have inspired the English tongue twister...**
 - a) "She sells seashells by the seashore"
 - b) "London Bridge Is Falling Down"
 - c) "Twinkle, Twinkle, Little Star"
 - d) "Humpty Dumpty"
2. 👁 **The first dinosaur was scientifically named in 1824. It was...**
 - a) Megalosaurus
 - b) Diplodocus
 - c) Tyrannosaurus
 - d) Velociraptor
3. 🗨 **In "Principles of Geology" (1830), Charles Lyell argued that the Earth was shaped by...**
 - a) slow processes still at work today
 - b) sudden miracles
 - c) humans
 - d) dinosaurs
4. 📅 **Dinosaurs died out about 66 million years ago, and the first ones appeared about 230 million years ago. For how many million years did they exist (not counting birds)?**

5. 🦕 **In 1980, Luis and Walter Alvarez linked the extinction of the dinosaurs to...**
 - a) an epidemic
 - b) the cold of an ice age
 - c) an asteroid impact
 - d) humans
6. 🌋 **The Chicxulub crater, left by an asteroid impact 66 million years ago, is in...**
 - a) Poland
 - b) Mexico
 - c) Australia
 - d) Egypt
7. 💖 **The age of the Earth – about 4.5 billion years – was determined in 1956 by...**
 - a) Charles Darwin
 - b) Isaac Newton
 - c) Albert Einstein
 - d) Clair Patterson
8. 🌱 **The first plants moved onto land about...**
 - a) 470 million years ago
 - b) 470 years ago
 - c) 4.7 billion years ago
 - d) 47,000 years ago
9. ⭐ **While studying lead, Clair Patterson also discovered that leaded gasoline...**
 - a) does not exist
 - b) is always cheaper
 - c) poisons people and the environment
 - d) is healthy

1 a 2. a 3. a 4. 1645. c 6. b 7. d 8. a 9. c

A N S W E R S

LINGUISTICS



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Let's travel back to the early history of the Earth and open the geologists' dictionary. Periods are named after places: "Jurassic" comes from the Jura Mountains on the border of France and Switzerland, and "Cambrian" from the Latin name for Wales. The name "Pangaea," meaning "all Earth," was given to the supercontinent by Alfred Wegener. A fossil is the remains of an organism that turned to stone or left an imprint in rock. Dinosaur species names usually consist of two Latin or Greek parts. "Tyrannosaurus rex" means king of the tyrant lizards, Velociraptor means "swift thief," and Triceratops means "three-horned face." The Polish dinosaur found in Lisowice was named Smok wawelski, the "Wawel Dragon." Remember these words – I'll ask you about them in a moment.

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1. 🏆 The name “Tyrannosaurus rex” means...
 - a) king of the tyrant lizards
 - b) old snake
 - c) great dragon
 - d) fast bird
2. 🗨 Geological periods are often named after places – the name “Jurassic” comes from mountains...
 - a) on the border of France and Switzerland
 - b) on the Baltic coast
 - c) in southern Poland
 - d) in the western USA
3. 📖 The name “Cambrian” comes from the Latin name for...
 - a) Wales
 - b) Italy
 - c) Poland
 - d) Egypt
4. 📏 The names of dinosaur species usually consist of...
 - a) only the discoverer’s name
 - b) two Latin or Greek parts
 - c) a single English word
 - d) a number
5. 🦖 Velociraptor means...
 - a) long neck
 - b) three horns
 - c) swift thief
 - d) great tail
6. 🐉 The Polish dinosaur-like predator found in Lisowice was named...
 - a) Smok wawelski (“Wawel Dragon”)
 - b) Krakus
 - c) Varsovius
 - d) Polonosaurus
7. 💖 Triceratops means...
 - a) three brothers
 - b) three-horned face
 - c) three legs
 - d) three tails
8. 🌿 A fossil is the remains of an organism that has...
 - a) frozen
 - b) turned to stone or left an imprint in rock
 - c) become plastic
 - d) stayed very old but soft
9. 🌟 The name “Pangaea”, meaning “all the Earth”, was given to the supercontinent by...
 - a) Charles Darwin
 - b) James Hutton
 - c) Mary Anning
 - d) Alfred Wegener

LITERARY STUDIES



2. Beginning of the Earth

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Let's travel back to the early history of the Earth – as writers imagined it. In Verne's "Journey to the Center of the Earth," the heroes meet prehistoric animals underground. In "The Lost World" of 1912, Arthur Conan Doyle placed dinosaurs on a plateau resembling a tepui in Venezuela. Michael Crichton wrote the novel "Jurassic Park" in 1990, in which dinosaur DNA was taken from mosquitoes trapped in amber. In children's tales and cartoons, dinosaurs often talk and have adventures. In Tolkien, the Ents are tree-like shepherds of the forests, forests older than people. In the "Just So Stories," Rudyard Kipling playfully explains how the elephant got its trunk. And stories about the end of the dinosaurs remind us that no species lasts forever. Remember these stories – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🚩 The novel “The Lost World” (1912), about dinosaurs on a plateau in South America, was written by...
 - a) Jules Verne
 - b) Mark Twain
 - c) Arthur Conan Doyle
 - d) H.G. Wells
2. 👁 In Verne’s “Journey to the Center of the Earth”, the heroes meet underground...
 - a) prehistoric animals
 - b) aliens
 - c) gnomes
 - d) robots
3. 🗨 The novel “Jurassic Park” was written in 1990 by...
 - a) J.K. Rowling
 - b) Stanisław Lem
 - c) Stephen King
 - d) Michael Crichton
4. 🦕 In “Jurassic Park”, dinosaur DNA was obtained from...
 - a) bones
 - b) feathers
 - c) eggs
 - d) mosquitoes trapped in amber
5. 🦖 In children’s stories and animated films, dinosaurs often...
 - a) are always dangerous
 - b) are robots
 - c) talk and have adventures
 - d) never appear
6. 🧝 In Tolkien’s books, the beings called Ents are...
 - a) dragons
 - b) dwarves
 - c) tree-like shepherds of the forests
 - d) dinosaurs
7. 💖 In his “Just So Stories”, Rudyard Kipling humorously explains, for example, how...
 - a) the elephant got its trunk
 - b) people got phones
 - c) the dinosaurs died out
 - d) water became wet
8. 🌿 In “The Lost World”, Arthur Conan Doyle placed dinosaurs on a plateau resembling...
 - a) a tepui in Venezuela
 - b) the Tatra Mountains
 - c) the Sahara
 - d) the Alps
9. ⭐ Stories about the end of the dinosaurs remind people that...
 - a) humans lived with dinosaurs
 - b) asteroids do not exist
 - c) dinosaurs will return
 - d) no species lasts forever

CULTURE AND RELIGION



2. Beginning of the Earth

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Let's travel back to the early history of the Earth and see how it is reflected in culture and faith. The ancient Greeks explained huge bones as the remains of giants and monsters, and legendary dragons may have been inspired by fossils that people found. In Disney's "Fantasia," Stravinsky's "The Rite of Spring" illustrated the formation of the Earth and the dinosaurs. Dinosaurs became icons thanks to films, toys and museums; the Museum of Evolution of the Polish Academy of Sciences is in the Palace of Culture and Science in Warsaw. Polish land remembers ancient times too: the rocks of the Holy Cross Mountains are about 500 million years old, and Baltic amber formed from tree resin about 40 million years ago. The Church holds that the theory of evolution does not contradict faith in creation, and Saint John Paul II said science and faith are two wings – one can study nature and believe in the Creator. Remember these details – I'll ask you about them in a moment.

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1. 🎵 In Disney's "Fantasia", Igor Stravinsky's "The Rite of Spring" illustrated...
 - a) Snow White
 - b) a battle of aliens
 - c) the formation of the Earth and the dinosaurs
 - d) Mickey Mouse on ice skates
2. 👁 When the ancient Greeks found the bones of huge animals, they explained them as the remains of...
 - a) cows
 - b) giants and monsters
 - c) fish
 - d) humans
3. 🗨 The dragons of legend in many cultures may have originated from...
 - a) fossils that people found
 - b) dreams
 - c) lightning
 - d) clouds
4. 🏰 The Catholic Church holds that the theory of evolution...
 - a) is forbidden
 - b) does not contradict faith in creation
 - c) does not exist
 - d) is a fairy tale
5. 🎬 Dinosaurs became a pop-culture icon thanks in part to...
 - a) films, toys and museums
 - b) Roman mosaics
 - c) the Bible
 - d) medieval chronicles
6. 🏛 The Museum of Evolution of the Polish Academy of Sciences' Institute of Paleobiology is located in...
 - a) the Palace of Culture and Science in Warsaw
 - b) Malbork Castle
 - c) the Wieliczka Salt Mine
 - d) Wawel Castle
7. 💎 Baltic amber formed from the resin of trees that grew about...
 - a) 40,000 years ago
 - b) 40 million years ago
 - c) 4 billion years ago
 - d) 400 years ago
8. 🌿 The Holy Cross Mountains are among the oldest mountains in Poland – their rocks are about...
 - a) 500 million years old
 - b) 5,000 years old
 - c) 500 years old
 - d) 5 million years old
9. ✨ St. John Paul II said that faith and reason are like two wings. For the history of the Earth, this means that...
 - a) you must choose only one of them
 - b) one can study nature and believe in the Creator
 - c) faith is unnecessary
 - d) science is forbidden to believers

ECONOMICS AND FINANCE



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the early history of the Earth, when its treasures were formed. Iron ore deposits formed about 2 billion years ago thanks to oxygen produced by cyanobacteria. Diamonds were born deep in the Earth through enormous pressure and temperature. Hard coal formed mainly in the Carboniferous period, and petroleum from the remains of tiny marine organisms. Fossil fuels are “nonrenewable” because they took millions of years to form. The salt of Wieliczka formed about 13.6 million years ago when a sea evaporated, and amber workers in Gdańsk have for centuries worked amber into jewelry. Today even fossils have a price: the Tyrannosaurus skeleton “Stan” sold in 2020 for about 32 million dollars. The fossil trade is controversial because valuable specimens are lost to science. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🏆 In 2020, the Tyrannosaurus skeleton “Stan” was sold at auction for about...
 - a) 32 million dollars
 - b) 32 thousand dollars
 - c) 32 dollars
 - d) 3 billion dollars
2. 👁 Hard coal, on which industry was built, formed mainly during the period called the...
 - a) Cretaceous
 - b) Carboniferous
 - c) Jurassic
 - d) Cambrian
3. 📖 Crude oil formed mainly from the remains of...
 - a) humans
 - b) tiny marine organisms
 - c) dinosaurs
 - d) only conifers
4. ⚖ We call fossil fuels “non-renewable” because they...
 - a) are cheap
 - b) took millions of years to form
 - c) are green
 - d) are easy to produce
5. ⚔ For centuries, amber craftsmen in Gdańsk have...
 - a) built ships
 - b) caught fish
 - c) mined coal
 - d) worked amber into jewelry
6. 🧑 The salt in Wieliczka formed about 13.6 million years ago, when there evaporated a...
 - a) glacier
 - b) freshwater lake
 - c) river like the Vistula
 - d) sea
7. 💎 Diamonds formed deep inside the Earth through...
 - a) human activity
 - b) rain
 - c) enormous pressure and temperature
 - d) the freezing of water
8. 🌿 Iron ore deposits formed about 2 billion years ago thanks to...
 - a) oxygen produced by cyanobacteria
 - b) humans
 - c) dinosaurs
 - d) volcanoes on the Moon
9. ⭐ The fossil trade is controversial because...
 - a) fossils are cheap
 - b) valuable specimens are lost to science
 - c) nobody wants them
 - d) they are banned everywhere

PSYCHOLOGY



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the early history of the Earth and see what happens in our minds when we think about it. Children often love dinosaurs, and psychologists say this passion develops curiosity and memory. Huge skeletons in museums evoke a feeling of wonder and smallness. Millions of years are hard to imagine, so timelines and comparisons help: if Earth's history fit into 24 hours, the dinosaurs would die out about 21 minutes before midnight. Fear of movie monsters is a reaction that once protected life. Digs teach patience and teamwork, collecting stones teaches you to observe nature, and visiting a museum with family builds shared memories. The extinction of the dinosaurs teaches that it is worth caring for the Earth and its inhabitants. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🦖 Children often love dinosaurs. Psychologists say that such a passion...
 - a) develops curiosity and memory
 - b) is harmful
 - c) is strange
 - d) does not matter
2. 👁 Huge skeletons in museums give visitors a sense of...
 - a) sleepiness
 - b) wonder and smallness
 - c) boredom
 - d) hunger
3. 🗨 People find it hard to imagine millions of years. What helps is...
 - a) dreams
 - b) timelines and comparisons
 - c) counting on your fingers
 - d) horoscopes
4. ⏳ If Earth's history were squeezed into 24 hours, the dinosaurs would die out about 21 minutes before midnight. How many minutes is that, approximately (whole number)?

5. 🧑🏫 Fossil-dig expeditions teach young people...
 - a) laziness
 - b) haste
 - c) loneliness
 - d) patience and teamwork
6. 👨👩👦 Visiting a natural history museum together as a family...
 - a) is forbidden for children
 - b) is always boring
 - c) is expensive everywhere
 - d) builds shared memories
7. 💖 Fear of the huge animals in movies is an example of...
 - a) an illness
 - b) laziness
 - c) a reaction that once protected life
 - d) an invention
8. 🌿 Collecting rocks and fossils is a hobby that...
 - a) is forbidden
 - b) is dangerous
 - c) achieves nothing
 - d) teaches you to observe nature
9. ⭐ The thought of the dinosaurs' extinction can be frightening, but it also teaches us that...
 - a) we must live in constant fear
 - b) everything is pointless
 - c) nothing at all depends on us
 - d) it is worth caring for the Earth and its inhabitants

EDUCATION



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the early history of the Earth and see how we learn about it. In geography class we study the table of geologic eras and periods; the order of the eras from oldest is Paleozoic, Mesozoic, Cenozoic. To show how fossils form, teachers press shells into clay or plaster. Dinosaur songs help children remember names and facts, and the JuraPark in Białtów, Krasiejów and Solec Kujawski teaches about dinosaurs and ancient life. Geological trails, for example in the Holy Cross Mountains, let you see ancient rocks in the field. Mary Anning could not study because she was a woman, yet she taught herself and shared her knowledge with scientists. Olympiads let students develop their passions and win university admission. And lessons about extinction teach that humans also affect nature. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 **Children's songs about dinosaurs help kids...**
 - a) fall asleep at school
 - b) learn to fly
 - c) do nothing
 - d) remember names and facts
2. 👁️ **The JuraParks in Białtów, Krasiejów and Solec Kujawski teach children about...**
 - a) cars
 - b) aliens
 - c) dinosaurs and ancient life
 - d) the Middle Ages
3. 🗨️ **In geography lessons we learn a table of...**
 - a) geologic eras and periods
 - b) chemical elements only
 - c) multiplication
 - d) spelling rules
4. 📅 **The order of the geologic eras, from oldest, is...**
 - a) Cenozoic, Paleozoic, Mesozoic
 - b) Mesozoic, Paleozoic, Cenozoic
 - c) Paleozoic, Mesozoic, Cenozoic
 - d) Cenozoic, Mesozoic, Paleozoic
5. 🛠️ **To show students how fossils form, teachers...**
 - a) dig holes in the sports field
 - b) boil bones
 - c) press shells into clay or plaster
 - d) paint stones
6. 👤 **Mary Anning, although she could not study at university because she was a woman,...**
 - a) knew nothing at all about bones or fossils
 - b) wrote only poetry
 - c) was a university professor
 - d) taught herself and shared her knowledge with scientists
7. 💖 **The Biology and Geography Olympiads allow students to...**
 - a) develop their passions and win university admission
 - b) earn millions
 - c) go on a free vacation abroad every year
 - d) skip school for a year
8. 🌿 **Geological trails, for example in the Holy Cross Mountains, let you...**
 - a) buy souvenirs
 - b) go fishing
 - c) see ancient rocks in the field
 - d) drive a car
9. ⭐ **By learning about the extinction of species, students understand that...**
 - a) dinosaurs will return
 - b) humans also affect nature
 - c) nothing ever changes
 - d) learning is not worth it

SOCIOLOGY

2. Beginning of the Earth

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Let's travel back to the early history of the Earth and see how it brings people together. The question of life's origin now unites scientists, philosophers and theologians. The Māori, Aboriginal Australians and Native Americans have their own stories about the origin of mountains, rivers and animals. People settled volcanic lands because the soil there is fertile. In some countries, debates about evolution in schools cause social conflicts – the 1925 “Monkey Trial” involved a teacher who taught evolution against state law. Today science festivals like the Science Picnic draw tens of thousands of families. Volunteers help paleontologists with digs and cleaning bones. Residents of places with discoveries, like Krasiejów, develop tourism and education. And your own collection of stones can grow into a lifelong passion. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🚩 **Science festivals in Poland (e.g., the Science Picnic) attract...**
 - a) only professors
 - b) tens of thousands of families
 - c) no one
 - d) only children
2. 👁 **Residents of places with paleontological discoveries, such as Krasiejów, often...**
 - a) sell the bones
 - b) hide the discoveries
 - c) do not know about them
 - d) develop tourism and education
3. 🗨 **In some countries, debates about teaching evolution in schools...**
 - a) cause social conflicts
 - b) are only about fish
 - c) never happen
 - d) are banned everywhere
4. 🐒 **The 1925 “Monkey Trial” concerned a teacher who...**
 - a) taught evolution against state law
 - b) did not come to school
 - c) wrote in Latin
 - d) stole a monkey
5. 🙋 **Volunteers help paleontologists with...**
 - a) cooking
 - b) building roads
 - c) digs and cleaning bones
 - d) only selling tickets
6. 🗣 **The Māori, Aboriginal Australians and Native Americans have their own stories about...**
 - a) factories and their machines
 - b) the origin of mountains, rivers and animals
 - c) computers
 - d) cars and motorcycles
7. 💖 **Your own collection of rocks or fossils can...**
 - a) grow into a lifelong passion
 - b) be harmful
 - c) be forbidden
 - d) always be boring
8. 🌿 **People have settled in volcanic areas because...**
 - a) it is cold
 - b) it is safe
 - c) there are no earthquakes
 - d) the soil there is fertile
9. 🌟 **Today, the question of the origin of life brings together...**
 - a) scientists, philosophers and theologians
 - b) only biologists
 - c) no one
 - d) only politicians

1. b 2. d 3. a 4. a 5. c 6. b 7. a 8. d 9. a

A N S W E R S

LAW



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the early history of the Earth and see how the law protects its traces. The 1972 UNESCO Convention protects natural sites such as the Grand Canyon and the Great Barrier Reef. The Grand Canyon exposes rocks up to 2 billion years old. The reef is protected because its destruction would be irreversible – reefs grow over thousands of years. UNESCO Geoparks protect exceptional geological heritage. In Mongolia fossils are protected by law, and they may not be exported without permission. In Poland, nature protection law provides for national parks and reserves; the Kadzielnia reserve in Kielce protects rocks and fossils. A natural monument of inanimate nature may be an exceptional rock or boulder. And anyone who finds a valuable fossil should report it to scientists. Remember these details – I'll ask you about them in a moment.

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Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **UNESCO Global Geoparks protect...**
 - a) musical monuments
 - b) old cinemas
 - c) exceptional geological heritage
 - d) airports
2. 👁️ **Fossils in Mongolia are protected by law, and...**
 - a) they always belong to the finder
 - b) they may not be exported without permission
 - c) they can be sold freely
 - d) they have no value
3. 🏠 **An example of an inanimate natural monument is...**
 - a) an old tree
 - b) a building
 - c) an exceptional rock or boulder
 - d) an animal
4. 🦋 **The Kadzielnia nature reserve in Kielce protects...**
 - a) gardens
 - b) horses
 - c) water birds
 - d) rocks and fossils
5. ✨ **Anyone in Poland who finds a valuable fossil should...**
 - a) destroy it
 - b) report it to scientists
 - c) sell it abroad
 - d) hide it at home
6. 🌍 **The UNESCO Convention (1972) protects natural sites such as...**
 - a) shopping malls
 - b) the parking lots of big cities
 - c) the Grand Canyon and the Great Barrier Reef
 - d) highways and bridges
7. 💜 **The Grand Canyon in the USA reveals rocks that are up to...**
 - a) 200 years old
 - b) 20 years old
 - c) 2 billion years old
 - d) 2,000 years old
8. 🌿 **Nature conservation law in Poland provides for forms of protection such as...**
 - a) national parks and reserves
 - b) airports
 - c) allotment gardens
 - d) only zoos
9. ☆ **The Great Barrier Reef is protected because...**
 - a) it is small
 - b) it is ugly
 - c) its destruction would be irreversible
 - d) it is artificial

OTHER (HUMANITIES)



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the early history of the Earth, written in its oldest stones. The oldest rocks, for example in Canada, are about 4 billion years old, and the oldest mineral grains – zircons from Australia – are about 4.4 billion years old. Mount Ślęza and the Sudetes are made of rocks formed hundreds of millions of years ago. Dinosaurs lived for a very long time: Tyrannosaurus lived closer in time to us than to Stegosaurus. Copies of their skeletons in museums are made of lightweight materials cast from molds. Dinosaurs are everywhere in culture: the band "T. Rex" took its name from the Tyrannosaurus, and the dinosaur in the Chrome browser appears when there is no internet connection. In Polish, "old as a dinosaur" means very outdated. And the image of Earth as a "pale blue dot" makes us reflect on how fragile our home is. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎸 The rock band T. Rex took its name from...
 - a) the Tyrannosaurus
 - b) a tank
 - c) an airplane
 - d) a train
2. 👁 The dinosaur from the Chrome browser game appears when...
 - a) the computer overheats
 - b) there is no internet connection
 - c) it is night
 - d) the battery dies
3. 🗨 In everyday speech, calling something “a dinosaur” means it is...
 - a) very outdated
 - b) very strong
 - c) fast
 - d) green
4. ⚖ Tyrannosaurus lived closer in time to us than to...
 - a) humans
 - b) cats
 - c) mammoths
 - d) Stegosaurus
5. 🦴 Replica dinosaur skeletons in museums are made of...
 - a) wood
 - b) lightweight materials cast from molds
 - c) ice
 - d) always real bones
6. 🌍 The oldest rocks on Earth, for example in Canada, are about...
 - a) 4 million years old
 - b) 40 years old
 - c) 4 billion years old
 - d) 400 years old
7. 💎 The oldest mineral grains on Earth – zircons from Australia – are about...
 - a) 44 years old
 - b) 440,000 years old
 - c) 4.4 billion years old
 - d) 4,400 years old
8. 🌿 Mount Ślęza and the Sudetes are made of rocks that formed...
 - a) in the Middle Ages
 - b) hundreds of millions of years ago
 - c) after the war
 - d) yesterday
9. ☆ The image of the Earth as a “pale blue dot” prompts us to...
 - a) start arguing
 - b) feel bored
 - c) reflect on how fragile our home is
 - d) go shopping

1.a.2.b.3.a.4.d.5.b.6.c.7.c.8.b.9.c

ANSWERS

ARCHITECTURE AND URBAN PLANNING

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young and meet its builders. Karst caves, such as Raj Cave, formed from water dissolving limestone. Coral reefs are built by tiny animals called corals. Termite mounds have a system of natural ventilation, and beaver dams are an example of animal construction. The limestone of many cathedrals formed from shells and skeletons of sea creatures. The Wieliczka mine was dug into salt deposits that formed from an evaporated sea, and St. Kinga's Chapel there is carved in rock salt. To protect a dinosaur skeleton for transport, paleontologists wrap it in plaster bandages. And engineers building in earthquake zones remember that tectonic plates are still moving. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏰 **Karst caves, such as the Raj Cave in Poland, were formed by...**
 - a) wind wearing down granite
 - b) the sun melting ice
 - c) rain washing away sand
 - d) water dissolving limestone
2. 👁 **Coral reefs are structures built by...**
 - a) volcanoes
 - b) fish
 - c) humans
 - d) tiny animals called corals
3. 🐜 **Termite mounds, built for millions of years, have a system of...**
 - a) gas heating
 - b) natural ventilation
 - c) elevators
 - d) windows
4. 🏰 **The Wieliczka Salt Mine was dug into deposits that formed from...**
 - a) an evaporated sea
 - b) a glacier
 - c) desert sand
 - d) lava
5. 🦖 **To protect a dinosaur skeleton for transport, paleontologists...**
 - a) freeze it
 - b) put it in water
 - c) tape it together
 - d) wrap it in plaster bandages
6. 🏰 **St. Kinga's Chapel in Wieliczka is carved out of...**
 - a) wood
 - b) ice
 - c) rock salt
 - d) marble
7. ❤ **Beaver dams are an example of...**
 - a) animal construction
 - b) pure chance
 - c) children's play
 - d) a human structure
8. 🌿 **Limestone, used to build many cathedrals, formed from...**
 - a) ice
 - b) desert sand
 - c) lava
 - d) shells and skeletons of sea creatures
9. ⭐ **When building in earthquake zones, engineers take into account that tectonic plates...**
 - a) stand still
 - b) are still moving
 - c) disappear
 - d) are made of ice



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and lightning struck it – lightning may have helped create molecules needed for life. Earth's magnetic field is generated in the liquid iron core. A compass works because the needle lines up with the Earth's magnetic field. The magnetic poles have swapped places many times, and rocks containing magnetite record the former direction of the magnetic field. Without this field, the solar wind would strip away the air. A seismograph records the Earth's vibrations, and the result can be turned into sound; a worldwide network of seismographs helps warn of earthquakes and tsunamis. And electric eels and electric rays evolved to hunt and defend themselves with electricity. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **A seismograph records the Earth's vibrations, and the result can be turned into...**
 - a) smell
 - b) taste
 - c) heat
 - d) sound
2. 👁 **The Earth's magnetic field is generated in...**
 - a) the crust
 - b) the liquid iron core
 - c) the atmosphere
 - d) the oceans
3. 🗺 **A compass works because the magnetic needle lines up with...**
 - a) the Earth's magnetic field
 - b) the Sun
 - c) the wind
 - d) rivers
4. ⚡ **Over Earth's history, the magnetic poles have...**
 - a) never changed
 - b) swapped places many times
 - c) disappeared forever
 - d) been located in Poland
5. ⚡ **Lightning striking the early Earth may have helped create...**
 - a) mountains
 - b) water
 - c) continents
 - d) molecules needed for life
6. 🌐 **A worldwide network of seismographs makes it possible to...**
 - a) forecast the weather
 - b) warn of earthquakes and tsunamis
 - c) catch fish
 - d) measure height
7. 💖 **Electric eels and electric rays evolved in order to...**
 - a) glow in the dark
 - b) fly
 - c) hunt and defend themselves with electricity
 - d) sing to each other at night
8. 🌿 **Rocks containing magnetite record the past direction of...**
 - a) the magnetic field
 - b) the wind
 - c) rivers
 - d) the sun
9. ⭐ **Without the Earth's magnetic field,...**
 - a) the solar wind would strip away the air
 - b) it would be warmer
 - c) there would be no Moon
 - d) there would be no night

COMPUTER SCIENCE AND TELECOMMUNICATIONS

**2. Beginning of the Earth**

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and read the information it recorded. The first organisms probably stored hereditary information in RNA. Today DNA is life's "code," written with four letters, and by comparing species' DNA scientists build the family tree of life. Ants have communicated with scents (pheromones) for millions of years, and whales talk across hundreds of kilometers because sound travels well in water. Computers help too: models show how continents moved over millions of years, simulations show how dinosaurs walked and ran, and CT scans let us look inside fossilized eggs without destroying them. Scientists share 3D fossil models through online databases. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 **Whales can communicate over hundreds of kilometers because sound...**
 - a) is too quiet
 - b) travels well in water
 - c) needs air
 - d) does not work in water
2. 👁 **Computer models show how the ancient continents...**
 - a) were flat
 - b) did not exist
 - c) fell from the sky
 - d) moved over millions of years
3. 🗨 **DNA is the “code” of life, written using...**
 - a) pictures
 - b) four letters
 - c) 26 letters
 - d) two digits
4. ➡ **By comparing the DNA of different species, scientists build...**
 - a) a shopping list
 - b) the family tree of life
 - c) a calendar
 - d) a road map
5. ➡ **CT scanning lets scientists look inside...**
 - a) mountains
 - b) fossilized eggs without destroying them
 - c) oceans
 - d) cars
6. 🧑 **Scientists around the world share 3D models of fossils through...**
 - a) paper letters
 - b) carrier pigeons
 - c) the telegraph
 - d) online databases
7. 💖 **Computer simulations help us understand how dinosaurs...**
 - a) sang
 - b) talked
 - c) wrote
 - d) walked and ran
8. 🌿 **For millions of years, ants have communicated using...**
 - a) light
 - b) writing
 - c) scents (pheromones)
 - d) radio sounds
9. ⭐ **The first organisms probably stored hereditary information in...**
 - a) computers
 - b) books
 - c) RNA
 - d) DNA right from the start

1.b2.d3.b4.b5.b6.d7.d8.c9.c

ANSWERS

BIOMEDICAL ENGINEERING



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and see how evolution designed bodies. About 375 million years ago, animals like Tiktaalik, which had leg-like fins, moved onto land. The oldest tetrapod tracks in the world were found in Poland – at Zachełmie in the Holy Cross Mountains. A fish's fins, a frog's legs and our hands are similar because they come from a common ancestor. Mammals' ear bones evolved from reptile jawbones, and the human eye and the octopus eye arose independently of each other. Biomimetics imitates solutions found in nature: shark skin inspired drag-reducing swimsuits, and sharks' ever-renewing teeth inspire work on regrowing teeth. The study of dinosaur diseases is paleopathology. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The tiny ear bones of mammals evolved from...
 - a) feathers
 - b) reptile jawbones
 - c) scales
 - d) fish teeth
2. 👁 The human eye and the octopus eye evolved...
 - a) from a common eye
 - b) from stone
 - c) independently of each other
 - d) ready-made at once
3. 🗨 Biomimetics is a science that...
 - a) studies stones
 - b) imitates solutions found in nature
 - c) designs clothes
 - d) treats animals
4. 🐟 A fish's fins, a frog's legs and our hands have a similar bone structure because they...
 - a) are made of plastic
 - b) are alike by chance
 - c) were made by a doctor
 - d) come from a common ancestor
5. 🦕 The first animals to move onto land about 375 million years ago, such as Tiktaalik, had...
 - a) hands with fingers like ours
 - b) leg-like fins
 - c) wheels
 - d) wings
6. 🧑 The world's oldest tetrapod tracks were discovered in Poland, at...
 - a) Gdańsk
 - b) Warsaw
 - c) the Tatra Mountains
 - d) Zachełmie in the Holy Cross Mountains
7. ❤ Sharks replace their teeth throughout their lives, which inspires researchers to...
 - a) make knives
 - b) build boats
 - c) breed sharks
 - d) work on regrowing teeth
8. 🦈 Shark skin inspired the design of...
 - a) hats
 - b) drag-reducing swimsuits
 - c) shoes
 - d) umbrellas
9. ☆ Studying dinosaur diseases from their bones is called...
 - a) cardiology
 - b) geology
 - c) paleopathology
 - d) pediatrics

CHEMICAL ENGINEERING



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and step into its chemistry lab. Volcanic gases formed the first atmosphere, still without oxygen. About 2.4 billion years ago the Great Oxidation was caused by cyanobacteria producing oxygen, and only once oxygen appeared could the ozone layer form. In photosynthesis, plants make sugar and oxygen from CO_2 and water. Hard coal formed from plants from swamp forests, amber is fossilized tree resin, and chalk comes from the skeletons of microscopic algae. Petrified wood forms when minerals replace the wood's tissues. And the great Permian extinction is linked to volcanic eruptions in Siberia. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Volcanic gases on the early Earth formed...**
 - a) forests
 - b) an ocean of milk
 - c) the first atmosphere
 - d) a meteor shower
2. 👁️ **The Great Oxidation Event (about 2.4 billion years ago) was caused by...**
 - a) volcanoes
 - b) humans
 - c) cyanobacteria producing oxygen
 - d) dinosaurs
3. 🗨️ **Photosynthesis is the process by which plants use CO₂ and water to produce...**
 - a) nitrogen
 - b) sugar and oxygen
 - c) salt and water
 - d) coal and fire
4. ⚖️ **The ozone layer could only form once the atmosphere contained...**
 - a) helium
 - b) oxygen
 - c) argon
 - d) hydrogen
5. 🪵 **Petrified wood forms when...**
 - a) wood freezes
 - b) wood burns
 - c) minerals replace the wood's tissues
 - d) someone paints it
6. 🌳 **Hard coal formed from...**
 - a) sand
 - b) dinosaurs
 - c) plants from swamp forests
 - d) lava
7. 💖 **Amber is fossilized...**
 - a) water
 - b) tree resin
 - c) bone
 - d) lava
8. 🍃 **The chalk we write with on a blackboard formed from...**
 - a) the skeletons of microscopic algae
 - b) ash
 - c) ice
 - d) salt
9. ⭐ **The Permian mass extinction (about 252 million years ago) is linked to...**
 - a) humans
 - b) dinosaurs
 - c) an impact in Poland
 - d) volcanic eruptions in Siberia

CIVIL ENGINEERING AND TRANSPORT

**2. Beginning of the Earth**

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and set off with its travelers. Continents drift at a few cm per year; the Himalayas formed when the Indian plate collided with Asia. About 3 million years ago the Isthmus of Panama allowed animals to cross between the continents. The first flying animals, insects, appeared about 400 million years ago; pterosaurs were flying reptiles, and birds descend from dinosaurs. Herds of dinosaurs traveled together, as parallel tracks of many individuals show. In Poland you can see dinosaur footprints in places such as Sołtyków and Bałtów. And a heavy sauropod like Brachiosaurus could weigh as much as several elephants. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🦋 **Pterosaurs were...**
 - a) swimming dinosaurs
 - b) flying reptiles
 - c) insects
 - d) birds
2. 👁 **The first flying animals, insects, appeared about...**
 - a) 400 million years ago
 - b) 40,000 years ago
 - c) 4 million years ago
 - d) 4 billion years ago
3. 🐣 **Birds are descended from...**
 - a) mammals
 - b) fish
 - c) insects
 - d) dinosaurs
4. ➡️ **Continents move at a rate of...**
 - a) not at all
 - b) a few cm per year
 - c) a meter per hour
 - d) a few km per day
5. 🦖 **Dinosaur footprints in Poland can be seen, for example, in...**
 - a) Zakopane
 - b) Łódź
 - c) Sottyków and Bałtów
 - d) Sopot
6. 🦖 **Dinosaur herds traveled together, as shown by...**
 - a) inscriptions
 - b) parallel tracks of many individuals
 - c) films
 - d) drawings
7. ❤ **Heavy sauropods such as Brachiosaurus could weigh as much as...**
 - a) a hundred people
 - b) several elephants
 - c) one cat
 - d) one car
8. 🌿 **The Himalayas formed when the Indian Plate...**
 - a) exploded
 - b) froze
 - c) sank
 - d) collided with Asia
9. ☆ **The land bridge between North and South America (the Isthmus of Panama) formed about 3 million years ago, which allowed...**
 - a) a canal to be built
 - b) animals to cross between the continents
 - c) the Atlantic to form
 - d) the dinosaurs to die out

MATERIALS ENGINEERING



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and examine its materials. Rocks are divided into igneous, sedimentary and metamorphic. Granite formed from magma that solidified deep underground, and marble is a metamorphic rock formed from limestone. Diamond and graphite are two forms of carbon, and the hardest natural mineral is diamond. Quartz crystals vibrate regularly, so they are used in watches. Rare earth metals formed in stars, and reached the crust with magma. Nature makes super-materials too: mother-of-pearl is tough because it is built of layers like bricks, and spider silk is stronger than steel – spiders have existed for about 400 million years. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Quartz crystals vibrate regularly, so they are used in...**
 - a) drums
 - b) trumpets
 - c) guitars
 - d) watches
2. 👁 **Granite is a rock that formed from...**
 - a) marine sediments
 - b) sand
 - c) magma that solidified deep underground
 - d) ice
3. 🗨 **Rocks are divided into igneous, sedimentary and...**
 - a) metal
 - b) glass
 - c) metamorphic
 - d) plastic
4. ⚡ **Marble is a metamorphic rock that formed from...**
 - a) granite
 - b) ice
 - c) coal
 - d) limestone
5. ✨ **Mother-of-pearl in shells is very tough because it...**
 - a) is soft
 - b) is built of layers like bricks
 - c) is made of metal
 - d) is hollow
6. 🕸 **Spider silk is stronger than steel of the same thickness. Spiders have existed for...**
 - a) 400 years
 - b) 4,000 years
 - c) about 400 million years
 - d) 40 years
7. 💎 **Diamond and graphite are two forms of...**
 - a) silicon
 - b) iron
 - c) carbon
 - d) gold
8. 🌿 **The hardest natural mineral is...**
 - a) talc
 - b) quartz
 - c) gypsum
 - d) diamond
9. ⭐ **Rare earth metals, needed for phones, were formed in...**
 - a) stars, and reached the crust with magma
 - b) oceans
 - c) plants
 - d) factories

1.d 2.c 3.c 4.d 5.b 6.c 7.c 8.d 9.a

A N S W E R S

MECHANICAL ENGINEERING



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and visit nature's mechanics workshop. Dinosaur jaws worked like levers, and Tyrannosaurus had a bite force of several tens of thousands of newtons. Dinosaur tails were used for keeping their balance, and the long necks of sauropods let them reach leaves without walking. Kangaroos hop thanks to tendons that work like springs. Crickets chirp by rubbing their wings together. The wings of birds and pterosaurs are two ways of solving the problem of flight, and engineers learn from birds' curved wingtips, called winglets. And a robot dinosaur in a museum moves thanks to motors and controllers. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Crickets make music by rubbing...**
 - a) their antennae
 - b) their wings together
 - c) their tails
 - d) their legs on the ground
2. 👁️ **Dinosaur jaws worked like...**
 - a) levers
 - b) springs
 - c) pumps
 - d) wheels
3. 🗨️ **Tyrannosaurus had one of the strongest bite forces ever – about...**
 - a) 100 newtons
 - b) 10 newtons
 - c) the same as a human
 - d) several tens of thousands of newtons
4. 🐢 **The long necks of sauropods allowed them to...**
 - a) fly
 - b) swim
 - c) reach leaves without walking
 - d) dig burrows
5. 🦘 **Kangaroos jump thanks to tendons that work like...**
 - a) magnets
 - b) ropes
 - c) springs
 - d) wheels
6. 🤖 **A robot dinosaur in a museum moves thanks to...**
 - a) the wind
 - b) magic
 - c) motors and controllers
 - d) people always hidden inside
7. 💖 **Dinosaurs used their tails, among other things, for...**
 - a) always swimming
 - b) flying
 - c) keeping their balance
 - d) eating
8. 🦋 **The wings of birds and pterosaurs are an example of...**
 - a) the same bones
 - b) pure chance
 - c) a mistake
 - d) solving the problem of flight
9. ⭐ **Engineers designing airplanes learn from birds, for example from the upturned wingtips called...**
 - a) blades
 - b) wheels
 - c) winglets
 - d) sails



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and look for its energy reserves. The name of the Carboniferous period comes from the word for coal, and miners find fossilized leaves and trunks in coal. Brown coal is younger – it formed about 10 to 50 million years ago. Gas shales formed from mud on the floors of ancient seas, and oil and gas are now found using sound waves in rocks. Peat bogs store huge amounts of carbon. Burning fossil fuels releases CO₂ that plants absorbed millions of years ago. Using up in 200 years what took hundreds of millions of years to form, we must look for renewable sources, such as geothermal energy, whose heat comes partly from the Earth's formation and the decay of elements. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 Today, oil and gas are searched for using...
 - a) sound waves in rocks
 - b) dogs
 - c) magnets in boots
 - d) fortune-tellers
2. 👁 Lignite (brown coal) is younger than hard coal and formed about...
 - a) 4 billion years ago
 - b) in the Middle Ages
 - c) 10 to 50 million years ago
 - d) yesterday
3. 🗨 The name of the Carboniferous period comes from a word meaning...
 - a) salt
 - b) water
 - c) coal
 - d) ice
4. 🔥 Burning fossil fuels releases CO₂ that plants...
 - a) never knew
 - b) ate
 - c) absorbed millions of years ago
 - d) produced yesterday
5. ⚡ Miners in coal mines sometimes find...
 - a) fossilized leaves and trunks
 - b) living dinosaurs
 - c) human bones
 - d) gold coins
6. 🌋 Geothermal energy uses the Earth's heat, which comes partly from...
 - a) only the Sun
 - b) the wind
 - c) its formation and the decay of elements
 - d) the Moon
7. 💖 Shale, from which gas is extracted, is a rock formed from...
 - a) desert sand
 - b) ice
 - c) mud on the floors of ancient seas
 - d) lava
8. 🍃 Peat bogs form over thousands of years and store...
 - a) salt water
 - b) huge amounts of carbon
 - c) gold
 - d) oil
9. ☆ Using up in 200 years fuels that took hundreds of millions of years to form means that...
 - a) reserves are infinite
 - b) the fuels will grow back tomorrow
 - c) nothing will happen
 - d) we must look for renewable sources

1.a,2.c,3.c,4.c,5.a,6.c,7.c,8.b,9.d

ANSWERS

ASTRONOMY



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and the sky looked completely different. The Moon most likely formed after the young Earth collided with a planet called Theia. The young Moon was closer, so it looked much larger in the sky; today it moves away by about 3.8 cm a year. A day on the young Earth was shorter – only a few hours long. Corals from 400 million years ago show that a year then had about 400 days. Tides are caused mainly by the Moon. During the Great Bombardment, the Earth and Moon were struck by countless asteroids, whose traces are impact craters. Earth's water may have come partly from comets and asteroids. And the asteroid that hit Earth 66 million years ago was about 10 km across. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The Moon most likely formed after the young Earth collided with a planet called...
 - a) Gaia
 - b) Nibiru
 - c) Vulcan
 - d) Theia
2. 👁 The young Moon was much closer to the Earth, so in the sky it looked...
 - a) invisible
 - b) green
 - c) much larger
 - d) smaller
3. 🗨 The Late Heavy Bombardment was a period when the Earth and Moon were...
 - a) struck by lightning
 - b) struck by countless asteroids
 - c) lit by a brighter Sun
 - d) covered in snow
4. ⚡ A day on the young Earth was...
 - a) the same length
 - b) eternal
 - c) longer – a week long
 - d) shorter – only a few hours long
5. ✨ Traces of asteroid impacts on Earth are...
 - a) impact craters
 - b) volcanoes
 - c) always lakes
 - d) rivers
6. 🧑 The water on Earth may have come partly from...
 - a) the Sun
 - b) the Moon
 - c) comets and asteroids
 - d) Mars
7. 💖 Growth layers in corals from 400 million years ago show that a year then had about...
 - a) 100 days
 - b) 365 days
 - c) 400 days
 - d) 1,000 days
8. 🍃 The tides on Earth are caused mainly by...
 - a) the Sun alone
 - b) the wind
 - c) Mars
 - d) the Moon
9. ⭐ The asteroid that hit the Earth 66 million years ago had a diameter of about...
 - a) 1 km
 - b) 10 km
 - c) 100 km
 - d) 10 m

MATHEMATICS



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and calculate its age. The principle of superposition says lower rock layers are older, and to estimate their age geologists measure layer thickness and the rate of deposition. Tree rings let us count years. Uranium-238 has a half-life of about 4.5 billion years, so it is suited to dating the oldest rocks. Each half-life leaves half: from 1,000 atoms 500 remain, and after two half-lives 250. Ammonite shells have the shape of a logarithmic spiral, and hexagonal basalt columns form as lava cools. Relationships between species are worked out with cladistics – family trees. And of all species that ever lived, over 99% have gone extinct. Remember these numbers – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The rhythm of tree-ring growth lets us count...**
 - a) days
 - b) seconds
 - c) years
 - d) hours
2. 👁 **Ammonite shells have the shape of a...**
 - a) square
 - b) triangle
 - c) logarithmic spiral
 - d) straight line
3. 🗨 **The half-life of uranium-238 is about 4.5 billion years, so it is suitable for...**
 - a) measuring height
 - b) dating the oldest rocks
 - c) counting money
 - d) cooking
4. ➡ **If 500 out of 1,000 atoms of an isotope remain after one half-life, how many will remain after two half-lives?**

5. ⚡ **To estimate the age of rock layers, geologists measure their thickness and the rate of...**
 - a) deposition
 - b) melting
 - c) glowing
 - d) tree growth
6. 🧑 **Paleontologists from different countries work out how species are related using...**
 - a) horoscopes
 - b) drawing lots
 - c) cladistics – family trees
 - d) opinion polls
7. 💖 **The principle of superposition in geology says that lower layers are...**
 - a) from another continent
 - b) the same age
 - c) older
 - d) younger
8. 🌿 **Hexagonal basalt columns, like those at the Giant's Causeway, form when...**
 - a) it rains
 - b) lava cools
 - c) the wind blows
 - d) ice melts
9. ☆ **Of all the species that have ever lived on Earth, it is estimated that...**
 - a) half live on Mars
 - b) over 99% have gone extinct
 - c) only 1% have gone extinct
 - d) all are still alive

1 c 2 c 3 b 4 2505 a 6 c 7 c 8 b 9 b

A N S W E R S

OTHER (TECHNOLOGY)



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and see how technology brings dinosaurs back to life. Paleoartists draw them based on bones, tracks and living animals. Discoveries in China since the 1990s showed that many dinosaurs had feathers, and their colors were worked out by studying melanosomes under a microscope. The sound of *Parasaurolophus* is recreated by modeling its hollow crest. To learn what a dinosaur ate, researchers study fossilized droppings – coprolites. 3D printers are used to print copies of bones, and virtual exhibitions mean anyone can view fossils. Satellites help point to places where there may be fossils. And will dinosaurs return? Scientists say DNA does not survive that long. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏆 **Reconstructions of dinosaur sounds, for example of Parasaurolophus, are made by...**
 - a) only listening to birds
 - b) guessing without any data
 - c) modeling its hollow crest
 - d) using recordings from the era
2. 👁 **Paleoartists draw dinosaurs based on...**
 - a) bones, tracks and living animals
 - b) photographs
 - c) films
 - d) fantasy
3. 🗨 **Museums now create virtual exhibitions, thanks to which...**
 - a) bones no longer need studying
 - b) anyone can view fossils
 - c) museums have closed
 - d) dinosaurs come to life
4. 🦖 **The feather color of some dinosaurs was determined by examining...**
 - a) birds in the zoo
 - b) melanosomes under a microscope
 - c) stones
 - d) children's drawings
5. 🖨 **In paleontology, 3D printers are used to...**
 - a) print copies of bones
 - b) only decorate museums
 - c) print newspapers
 - d) print living dinosaurs
6. 🧑 **Dinosaur discoveries in China since the 1990s have shown that many of them had...**
 - a) metal scales
 - b) bat-like wings
 - c) six legs
 - d) feathers
7. ❤ **To find out what a dinosaur ate, researchers analyze...**
 - a) photographs
 - b) fossilized droppings – coprolites
 - c) feathers
 - d) recordings
8. 🌿 **Satellites help geologists identify places where...**
 - a) there may be fossils
 - b) there are no people
 - c) there are no rocks
 - d) there is always gold
9. ⭐ **Can dinosaurs be brought back from DNA? Scientists say that...**
 - a) DNA is in every stone
 - b) it is easy
 - c) DNA does not survive that long
 - d) it has already been done

1. c 2. a 3. b 4. b 5. a 6. d 7. b 8. a 9. c

ANSWERS

BIOLOGY

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young and life was just awakening. The oldest known traces of life are about 3.5–3.7 billion years old. Stromatolites, among the oldest traces, were built by cyanobacteria (blue-green algae). About 540 million years ago came the Cambrian explosion, when many new groups of animals appeared, including trilobites – extinct marine arthropods. The first animals to move onto land were fish-like tetrapods, and the first animals to make sounds on land were probably insects. Lichens are a partnership of a fungus and an alga. The first mammals lived alongside the dinosaurs and were small, about the size of a mouse. The first flowers appeared about 130 million years ago. The dinosaurs, except for birds, died out about 66 million years ago. And we know of five great extinctions in Earth's history. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The first animals to make sounds on land were probably...**
 - a) mammals
 - b) insects
 - c) fish
 - d) birds
2. 👁 **The Cambrian explosion (about 540 million years ago) was a time when...**
 - a) humans appeared
 - b) the dinosaurs died out
 - c) the Moon formed
 - d) many new groups of animals appeared
3. 📖 **Trilobites were extinct...**
 - a) marine arthropods
 - b) dinosaurs
 - c) mammals
 - d) birds
4. ⚖ **The oldest known traces of life on Earth are about...**
 - a) 3.5–3.7 billion years old
 - b) 35 million years old
 - c) 5 billion years old
 - d) 350 years old
5. 🦎 **The first animals to move from water onto land were...**
 - a) birds
 - b) fish-like tetrapods
 - c) dinosaurs
 - d) mammals
6. 🧑‍🤝🧑 **Lichens are an example of cooperation between...**
 - a) a fish and a bird
 - b) a fungus and an alga
 - c) a dog and a cat
 - d) a tree and a stone
7. ❤ **The first mammals lived alongside the dinosaurs and were...**
 - a) enormous
 - b) only aquatic
 - c) small, about the size of a mouse
 - d) flying
8. 🌿 **Which organisms built stromatolites, some of the oldest traces of life on Earth?**
 - a) Corals
 - b) Cyanobacteria (blue-green algae)
 - c) Ferns
 - d) Trilobites
9. ☆ **How many mass extinctions do we know of in Earth's history?**
 - a) none
 - b) one
 - c) five
 - d) ten

1.b 2.d 3.a 4.a 5.b 6.b 7.c 8.b 9.c

ANSWERS

CHEMISTRY

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young and look into its chemistry. The early atmosphere contained a lot of carbon dioxide and water vapor. When cyanobacteria began producing oxygen, it reacted with iron in the oceans, forming banded iron ores. Today oxygen makes up about 21% of the air. Carbon compounds, on which life is based, are called organic. The 1953 Miller-Urey experiment showed that gases and sparks can produce amino acids, and hydrothermal vents on the ocean floor may have been where life began. The ocean is salty because for millions of years rivers washed salts out of rocks. Too much CO₂ in the oceans once caused acidification and die-offs of organisms. And a layer rich in iridium points to an asteroid impact. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 The Earth's early atmosphere contained a lot of...
 - a) carbon dioxide and water vapor
 - b) perfume
 - c) oxygen
 - d) ozone
2. 👁 Banded iron formations were created when oxygen from cyanobacteria reacted with iron in the...
 - a) oceans
 - b) volcanoes
 - c) ice
 - d) air
3. 📖 The carbon compounds on which life is based are called...
 - a) gaseous
 - b) inorganic
 - c) metallic
 - d) organic
4. ⚡ The Miller–Urey experiment (1953) showed that gases and sparks can produce...
 - a) dinosaurs
 - b) amino acids
 - c) salts
 - d) stones
5. 🔥 Hydrothermal vents on the ocean floor may have been the place where...
 - a) dinosaurs lived
 - b) life began
 - c) bombs exploded
 - d) continents formed
6. 🧑 The iridium-rich layer at the Cretaceous–Paleogene boundary points to...
 - a) humans
 - b) rain
 - c) an asteroid impact
 - d) only a volcanic eruption
7. ❤ Today, the share of oxygen in the air is about...
 - a) 78%
 - b) 21%
 - c) 50%
 - d) 1%
8. 🌿 The ocean is salty because, over millions of years, rivers...
 - a) washed salts out of rocks
 - b) froze
 - c) poured salt from the sky
 - d) evaporated
9. ⭐ In the past, too much CO₂ in the oceans caused...
 - a) the seas to freeze
 - b) fresh water
 - c) more fish
 - d) acidification and die-offs of organisms

PHYSICS



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and restless. Its interior is divided into crust, mantle and core. Seismic waves help us learn the structure of the Earth's interior – in 1936 the Danish seismologist Inge Lehmann used them to discover that the Earth has a solid inner core. Tectonic plates move thanks to movements of the hot mantle, and mountains form mainly through collisions of tectonic plates. Volcanoes erupt when magma and gases burst out to the surface. The Krakatoa eruption was heard about 4,800 km away – sound in air travels about 340 m/s. Earthquakes are measured on the magnitude scale (e.g., Richter). And a tsunami most often follows an undersea earthquake. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎐 The eruption of Krakatoa was heard about 4,800 km away. Sound travels through air at about...
 - a) 3 m/s
 - b) 340 km/s
 - c) 34 km/h
 - d) 340 m/s
2. 👁 The Earth's interior is divided into the crust, the mantle and the...
 - a) ocean
 - b) sun
 - c) atmosphere
 - d) core
3. 🗨 Seismic waves help us learn about...
 - a) the composition of the air
 - b) people's ages
 - c) the weather
 - d) the structure of the Earth's interior
4. 🔄 Tectonic plates move thanks to...
 - a) tides
 - b) movements of the hot mantle
 - c) winds
 - d) humans
5. 📏 Earthquakes are measured on a scale of...
 - a) Mohs
 - b) Celsius
 - c) Beaufort
 - d) magnitude (e.g., Richter)
6. 🧑🏫 In 1936, Danish seismologist Inge Lehmann discovered that...
 - a) there is no core
 - b) the Earth is flat
 - c) the Moon is hollow
 - d) the Earth has a solid inner core
7. 💖 Volcanoes erupt when...
 - a) magma and gases burst out
 - b) heavy rain falls
 - c) winter arrives
 - d) a strong wind blows
8. 🌿 Mountains are formed mainly by...
 - a) wind
 - b) rain
 - c) humans
 - d) collisions of tectonic plates
9. ⭐ A tsunami most often forms after...
 - a) rain
 - b) a storm
 - c) an undersea earthquake
 - d) a high tide

ENVIRONMENT

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young and see its changing environment. About 700 million years ago, according to the “Snowball Earth” hypothesis, almost the entire Earth was covered in ice. The oldest forests appeared about 385 million years ago, and Carboniferous forests were full of giant ferns, horsetails and insects. Pangaea was a supercontinent joining almost all land; as it broke apart, the Atlantic Ocean formed. In the Permian and Triassic, the lands of Poland were often hot and dry, and where the Tatras stand today there was a sea. In the age of dinosaurs the climate was generally warmer than today. An ecosystem is a community of organisms and their environment, and the great extinctions teach that rapid climate change threatens many species. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 The forests of the Carboniferous were full of...
 - a) oaks
 - b) grasses
 - c) palms and cacti
 - d) giant ferns, horsetails and insects
2. 👁️ "Snowball Earth" is the hypothesis that about 700 million years ago...
 - a) there was no water
 - b) it was one big desert
 - c) almost the entire Earth was covered in ice
 - d) the Earth was extremely hot
3. 🗨️ The climate in the age of the dinosaurs was generally...
 - a) warmer than today
 - b) colder
 - c) icy
 - d) the same
4. 🏔️ Where the Tatra Mountains stand today, long ago there was...
 - a) a rainforest
 - b) a sea
 - c) an ice desert
 - d) a volcano
5. ✂️ When Pangaea broke apart, what formed was the...
 - a) Black Sea
 - b) Baltic Sea
 - c) Pacific Ocean
 - d) Atlantic Ocean
6. 🌿 An ecosystem is...
 - a) a single animal
 - b) a single species of plant
 - c) a stone
 - d) a community of organisms and their environment
7. 💜 During the Permian and Triassic, the area of present-day Poland was often...
 - a) under a deep ocean
 - b) hot and dry
 - c) a jungle
 - d) covered in ice
8. 🌿 The oldest forests on Earth appeared about...
 - a) 38,000 years ago
 - b) 385 million years ago
 - c) 3.8 billion years ago
 - d) 385 years ago
9. ☆ Mass extinctions teach us that rapid climate change...
 - a) is made up
 - b) threatens many species
 - c) does not matter
 - d) helps everyone

HEALTH

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young and discover where our health comes from. Bacteria have lived on Earth for about 3.5 billion years, so they are everywhere. The mitochondria in our cells were once free-living bacteria. Our spine is an inheritance from ancient fish, and our immune system comes from mechanisms that appeared in ancient invertebrates and fish. Mammals have three tiny bones in the middle ear, which gives them better hearing. The biological clock adapted to a day that was once shorter. The oxygen we breathe is produced mainly by ocean algae and plants. Paleontologists have found signs of fractures, infections and tumors in dinosaurs, and such diseases in fossils help us today understand where diseases come from. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🦋 **Mammals have a middle ear with three tiny bones, which gives them...**
 - a) better eyesight
 - b) wings
 - c) scales
 - d) better hearing
2. 👁 **Paleontologists have found evidence in dinosaurs of...**
 - a) winter colds
 - b) cavities from eating candy
 - c) fractures, infections and tumors
 - d) illnesses caused by phones
3. 🗨 **Our immune system derives from mechanisms that first appeared...**
 - a) in medieval humans
 - b) in the 20th century
 - c) in ancient invertebrates and fish
 - d) yesterday
4. ➡ **Bacteria have lived on Earth for about 3.5 billion years, which is why they...**
 - a) are everywhere
 - b) are rare
 - c) have died out
 - d) do not matter
5. 🦋 **The human spine is an inheritance from...**
 - a) insects
 - b) ancient fish
 - c) plants
 - d) snails
6. 👤 **The mitochondria in our cells were once...**
 - a) crystals
 - b) snails
 - c) free-living bacteria
 - d) viruses
7. ❤ **The biological clocks of organisms adapted to a day that was once...**
 - a) eternal
 - b) always the same
 - c) longer
 - d) shorter
8. 🌿 **The oxygen we breathe is produced mainly by...**
 - a) ocean algae and plants
 - b) rocks
 - c) cars
 - d) animals
9. ☆ **Diseases discovered in fossils help us today to...**
 - a) do nothing
 - b) ban diseases
 - c) treat dinosaurs
 - d) understand where diseases come from

PHYSICAL CULTURE AND SPORT

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young and watch a prehistoric competition. The fastest dinosaur was probably Ornithomimus. Reptiles such as ichthyosaurs and plesiosaurs swam in the seas. The Carboniferous dragonfly Meganeura had a wingspan of about 70 cm because the air held more oxygen than today. Turtles have existed for over 200 million years thanks to a strategy of a shell and slowness. Today the cheetah reaches about 100 km/h, kangaroos move by hopping on their hind legs, and flying fish glide over the water to escape predators. Pack hunting, as in wolves, requires cooperation within the group. And evolution is not a race to a finish line but adaptation to conditions. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏆 **The fastest dinosaur was probably...**
 - a) Brachiosaurus
 - b) Ankylosaurus
 - c) Stegosaurus
 - d) Ornithomimus
2. 👁 **The cheetah, the fastest land animal, reaches about...**
 - a) 100 km/h
 - b) 10 km/h
 - c) 40 km/h
 - d) 300 km/h
3. 🐡 **Swimming reptiles from the age of the dinosaurs include...**
 - a) ichthyosaurs and plesiosaurs
 - b) mammoths
 - c) tyrannosaurs
 - d) pterosaurs
4. 🦋 **The dragonfly Meganeura (Carboniferous) had a wingspan of about 70 cm because the air contained...**
 - a) less oxygen
 - b) more oxygen than today
 - c) hydrogen
 - d) more helium
5. 🦘 **Kangaroos get around by...**
 - a) hopping on their hind legs
 - b) crawling
 - c) flying
 - d) swimming
6. 🐺 **Hunting in packs, as wolves do, requires...**
 - a) luck
 - b) noise
 - c) solitude
 - d) cooperation within the group
7. 🐢 **Turtles have existed for more than 200 million years, and their survival strategy is...**
 - a) venom
 - b) flight
 - c) speed
 - d) a shell and slowness
8. 🐟 **Flying fish glide above the water in order to...**
 - a) escape predators
 - b) get a breath of air
 - c) win competitions
 - d) catch birds
9. ⭐ **Evolution is not a race toward a goal, but rather...**
 - a) a competition for a medal
 - b) a human plan
 - c) adaptation to conditions
 - d) chance without any rules

MEDICINE

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young – many of today's medicines have roots in that evolution. Fungi and bacteria have fought each other for millions of years, so antibiotics such as penicillin come from fungi and bacteria. Bacteria developed resistance to antibiotics before humans discovered them. The partnership between bacteria and humans in our gut has roots in the ancient evolution of animals. Snake and scorpion venom is used today in making medicines, and leeches help because they prevent blood from clotting. Sharks, resistant to many diseases, are studied by scientists looking for new medicines. Dolphins and bats “see” with sound thanks to echolocation – ultrasound works in a similar way. The oldest fossil tumor in a dinosaur was found in a leg bone. And knowledge of how viruses evolve helps doctors prepare vaccines. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🦋 Dolphins and bats “see” with sound thanks to...
 - a) echolocation
 - b) their eyes
 - c) smell
 - d) magnets
2. 👁️ The oldest fossil evidence of a cancerous tumor in a dinosaur was found in...
 - a) an egg
 - b) a tooth
 - c) a feather
 - d) a leg bone
3. 📖 Antibiotics, such as penicillin, come from...
 - a) stones
 - b) humans
 - c) fungi and bacteria
 - d) fish
4. ⚖️ Bacteria developed resistance to antibiotics...
 - a) only in 2000
 - b) before humans discovered them
 - c) never
 - d) in the Middle Ages
5. 🦈 Sharks are resistant to many diseases, which is studied by...
 - a) cooks
 - b) divers
 - c) scientists looking for new medicines
 - d) fishermen
6. 🧬 The symbiosis between humans and the bacteria in our gut has its roots in...
 - a) the ancient evolution of animals
 - b) the Stone Age
 - c) the 20th century
 - d) the Middle Ages
7. 💖 The venom of snakes and scorpions, which evolved for defense and hunting, is used today in...
 - a) fuel
 - b) paints
 - c) cosmetics only
 - d) making medicines
8. 🐞 Medicinal leeches, which have existed for millions of years, help in medicine because they...
 - a) glow
 - b) prevent blood from clotting
 - c) eat bacteria
 - d) heal fractures
9. ⭐ Knowledge of how viruses evolve helps doctors...
 - a) stop evolution
 - b) prepare vaccines
 - c) breed dinosaurs
 - d) treat stones

PHARMACY

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young and visit nature's pharmacy. Plants make medicinal substances to defend themselves against animals and disease. Yew bark gave us a cancer drug, and the venom of the cone snail a drug for pain relief. The ginkgo, a "living fossil," has existed for about 200 million years. Field horsetail, a relative of the giant Carboniferous horsetails, is now a medicinal plant. Sponges, among the oldest animals, appeared about 600 million years ago, and drugs from them are studied by scientists from many countries working together. To discover new medicines, researchers study organisms from forests and oceans. That is why the extinction of plant species means we may lose undiscovered medicines. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🐌 The cone snail produces a venom that has been turned into a drug for...
 - a) pain relief
 - b) helping children sleep
 - c) appetite
 - d) a runny nose
2. 👁 The bark of the yew tree, a plant that has existed for millions of years, gave us a drug that treats...
 - a) cancer
 - b) toothache
 - c) flu
 - d) coughs
3. 🗨 The ginkgo tree, a “living fossil”, has existed for about...
 - a) 2 billion years
 - b) 2,000 years
 - c) 200 years
 - d) 200 million years
4. ⚖ Plants produce medicinal substances mainly in order to...
 - a) make money for farmers
 - b) smell nice to the people nearby
 - c) glow in the dark
 - d) defend themselves against animals and disease
5. 🧪 To discover new medicines, scientists study...
 - a) cars
 - b) organisms from forests and oceans
 - c) clouds
 - d) only stones
6. 🧑‍🔬 Drugs from marine organisms, such as sponges, are studied by scientists from...
 - a) many countries working together
 - b) the military only
 - c) a single city
 - d) Poland alone
7. ❤ Sponges are among the oldest animals, having appeared about...
 - a) 6 billion years ago
 - b) 60 years ago
 - c) 6 million years ago
 - d) 600 million years ago
8. 🌿 Field horsetail, a relative of the giant horsetails of the Carboniferous, is today...
 - a) a tree
 - b) always a poisonous weed
 - c) a medicinal plant
 - d) a fungus
9. ☆ The extinction of plant species means we could lose...
 - a) stones
 - b) undiscovered medicines
 - c) nothing
 - d) only decorations

AGRICULTURE

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young and see how its fertility was born. Soil forms through the weathering of rocks and decay of organisms. Chernozem, the most fertile soil, formed on grassy steppes. Grasses, the ancestors of cereals, spread several tens of millions of years ago. Bees and flowers have evolved together for about 100 million years. Nitrogen from the air is captured by bacteria in the roots of legumes, and earthworms loosen and enrich the soil. Insects invented farming too: leafcutter ants have grown fungi for about 50 million years, and termites raise them in their mounds. But soil is fragile – erosion is fast when forests are cut down and land is over-plowed. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🐝 **Bees and flowers have evolved together for about...**
 - a) 100 million years
 - b) 1 billion years
 - c) 10,000 years
 - d) 100 years
2. 🌾 **Grasses, from which cereal crops come, spread about...**
 - a) in the Middle Ages
 - b) several tens of millions of years ago
 - c) 4 billion years ago
 - d) yesterday
3. 🗑️ **Soil is formed by...**
 - a) the work of humans alone
 - b) factories and machines
 - c) meteor showers
 - d) weathering of rocks and decay of organisms
4. 🌱 **Plants need nitrogen, which is “captured” from the air by...**
 - a) flowers
 - b) bacteria in the roots of legumes
 - c) bees
 - d) leaves
5. 🪱 **Earthworms, which have lived for millions of years, help farmers because they...**
 - a) loosen and enrich the soil
 - b) eat the crops
 - c) glow
 - d) scare birds
6. 🐜 **Leafcutter (fungus-farming) ants have been growing fungi for about...**
 - a) 5,000 years
 - b) 500 years
 - c) 50 years
 - d) 50 million years
7. 💖 **Termites also “farm” fungi in their mounds, which shows that...**
 - a) fungi are animals
 - b) insects are lazy
 - c) termites eat stones
 - d) insects invented farming too
8. 🌿 **Chernozem (black earth), the most fertile soil, formed on...**
 - a) deserts
 - b) rocks
 - c) grassy steppes
 - d) glaciers
9. ⭐ **Soil erosion progresses quickly when...**
 - a) light rain falls
 - b) trees are planted
 - c) forests are cut down and land is over-plowed
 - d) grass grows thickly on it

VETERINARY SCIENCE

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young and step into a veterinarian's office. Vets know that birds are descendants of dinosaurs, and even chickens carry traces of dinosaurs in their genes. Songbirds have a voice organ called the syrinx. Crocodiles have existed for about 200 million years, and New Zealand's tuatara resembles reptiles from 200 million years ago. Birds, crocodiles, turtles and mammals survived the extinction of the dinosaurs. Reptiles, which need warmth from their surroundings, are called cold-blooded (ectothermic). Mammals and birds are warm-blooded, which lets them live in cold climates. And caring for endangered species matters because an extinct species cannot be brought back. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🦅 **Songbirds have a vocal organ called the...**
 - a) trumpet
 - b) syrinx
 - c) feather
 - d) beak
2. 👁 **Veterinarians who treat birds know that birds are descendants of...**
 - a) dinosaurs
 - b) mammals
 - c) fish
 - d) insects
3. 🐊 **Crocodiles have existed for about...**
 - a) 200 years
 - b) 2 billion years
 - c) 2 million years
 - d) 200 million years
4. 🐔 **Domestic chickens carry traces in their genes of ancestors such as...**
 - a) dinosaurs
 - b) bats
 - c) snakes
 - d) flying fish
5. 🌡 **Animals that depend on heat from their surroundings, such as reptiles, are called...**
 - a) cold-blooded (ectothermic)
 - b) aquatic
 - c) frozen
 - d) warm-blooded
6. 🐾 **Mammals and birds are warm-blooded, which allows them to...**
 - a) always fly
 - b) go without eating
 - c) live in cold climates
 - d) live without sleep
7. ❤ **The animals that survived the extinction of the dinosaurs include...**
 - a) birds, crocodiles, turtles and mammals
 - b) tyrannosaurs
 - c) every species
 - d) pterosaurs
8. 🦎 **The tuatara of New Zealand is a “living fossil” – it resembles reptiles from...**
 - a) 200 million years ago
 - b) 200 years ago
 - c) 2,000 years ago
 - d) 20 years ago
9. ☆ **Caring for endangered species is important because...**
 - a) all species live in zoos
 - b) species will restore themselves
 - c) it does not matter
 - d) an extinct species cannot be brought back

ANIMAL SCIENCE AND FISHERIES



2. Beginning of the Earth

4.5 bn years ago – 5 m years ago

Let's travel back to the days when the Earth was young and dive into ancient waters. Oysters and clams have lived on Earth for about 500 million years, and the nautilus resembles ancient ammonites. Sharks survived all five great extinctions because they are well adapted and diverse; the giant megalodon reached more than ten meters in length. Fish don't sink because they have a swim bladder. Lobe-finned fish, ancestors of four-legged animals, survive today as coelacanths and lungfish, and lungfish can breathe air and survive droughts. On land, horses evolved from a small animal the size of a dog. And honeybees have lived in colonies for several tens of millions of years. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Lobe-finned fish, the ancestors of tetrapods, survive today as...**
 - a) sharks
 - b) carp
 - c) salmon
 - d) coelacanths and lungfish
2. 🐙 **The nautilus resembles...**
 - a) ancient ammonites
 - b) crabs
 - c) land snails
 - d) fish
3. 🐚 **Oysters and clams have lived on Earth for about...**
 - a) 5 billion years
 - b) 5,000 years
 - c) 500 million years
 - d) 5 years
4. 🦈 **Sharks survived all five mass extinctions because they...**
 - a) are small
 - b) have armor
 - c) are well adapted and diverse
 - d) live on land
5. 🐟 **Fish do not sink because they have...**
 - a) a swim bladder
 - b) a balloon
 - c) wings
 - d) legs
6. 🐝 **Honeybees have lived in colonies for about...**
 - a) 100 years
 - b) 1,000 years
 - c) several tens of millions of years
 - d) 1 billion years
7. 🐾 **Horses evolved from a small animal about the size of a...**
 - a) mouse
 - b) elephant
 - c) dog
 - d) giraffe
8. 🦖 **Megalodon, a giant shark, reached a length of up to...**
 - a) 5 cm
 - b) 1 meter
 - c) more than ten meters
 - d) 100 meters
9. ⭐ **Lungfish are able to...**
 - a) breathe air and survive droughts
 - b) fly
 - c) live in ice
 - d) talk

1.d 2.a 3.c 4.c 5.a 6.c 7.c 8.c 9.a

ANSWERS

OTHER (NATURE)

2. Beginning of the Earth

4.5 bn years ago – 5 m years ago



Let's travel back to the days when the Earth was young and take a walk through its nature. The oldest known rock in Poland is about 2 billion years old. The Himalayas have marine fossils on their peaks because those rocks formed on an ancient ocean floor. The Galápagos Islands were formed by volcanoes. Australia's animals are unique because the continent was isolated for a long time. The name "dinosaur" does not include pterosaurs or marine reptiles. The first primates appeared about 55–65 million years ago, and bats about 50 million years ago – they "hear" the world through echolocation. Whales evolved from mammals that lived on land. And Earth's history teaches that humans are a very young species. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🦇 **Bats appeared about 50 million years ago and “hear” the world thanks to...**
 - a) echolocation
 - b) magnets
 - c) their eyes
 - d) smells
2. 👁 **The first primates, ancestors of monkeys and apes, appeared about...**
 - a) 550 years ago
 - b) 55–65 million years ago
 - c) 5 billion years ago
 - d) 5 years ago
3. 🗨 **The term “dinosaur” does not include pterosaurs or...**
 - a) Triceratops
 - b) Stegosaurus
 - c) Tyrannosaurus
 - d) marine reptiles
4. 🐋 **Whales evolved from mammals that lived...**
 - a) as fish
 - b) on land
 - c) as insects
 - d) as birds
5. 🗡 **The oldest known rock in Poland is about...**
 - a) 2 billion years old
 - b) 2,000 years old
 - c) 20 million years old
 - d) 200 years old
6. 🇦🇺 **Australia’s animals, such as kangaroos and koalas, differ from others because the continent...**
 - a) has no plants
 - b) has no sun
 - c) was isolated for a long time
 - d) is under water
7. 💖 **The Galápagos Islands were formed by...**
 - a) a continent
 - b) volcanoes
 - c) glaciers
 - d) a reef
8. 🌿 **The Himalayas have marine fossils on their peaks because...**
 - a) they fell from the sky
 - b) birds brought them
 - c) those rocks formed on an ancient ocean floor
 - d) people carried them up there
9. ⭐ **The history of the Earth teaches us that humans are...**
 - a) a very young species
 - b) the oldest species
 - c) an eternal species
 - d) the only species

1. a 2. b 3. d 4. b 5. a 6. c 7. b 8. c 9. a

ANSWERS